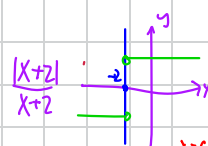


Section 2.4 (Continue)

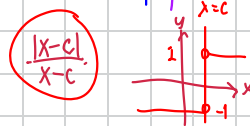
$$17.a) \lim_{x \rightarrow -2^+} (x+3) \frac{|x+2|}{x+2} = \lim_{x \rightarrow -2^+} (x+3)(1) = (-2+3)(1) = 1 \times 1 = 1$$



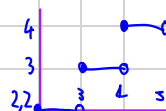
$$b) \lim_{x \rightarrow -2^-} (x+3) \frac{|x+2|}{x+2} = \lim_{x \rightarrow -2^-} (x+3)(-1) = (-2+3)(-1) = 1 \times (-1) = -1$$

$$20.a) \lim_{t \rightarrow 4^+} (t - |t|) = \lim_{t \rightarrow 4^+} t - 4 = 0$$

$$f(6) = 6 \\ f(6.7) = 6 \\ f(6.5) = 0$$



$$b) \lim_{t \rightarrow 4^-} (t - |t|) = \lim_{t \rightarrow 4^-} t - 3 = 1$$



$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = \frac{1}{1} = 1$$

$$28. \lim_{x \rightarrow 0} 6x^2 \cot x \csc 2x = \lim_{x \rightarrow 0} \frac{6x^2 \cdot \frac{\cos x}{\sin x} \cdot \frac{1}{2 \sin x \cos x}}{1} = \lim_{x \rightarrow 0} \frac{3x^2}{\sin^2 x} = 3 \cdot \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} = 3 \cdot \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} \right)^2 = 3 \cdot \left(\frac{1}{1} \right)^2 = 3$$

$$32. \lim_{x \rightarrow 0} \frac{x - x \cos x}{\sin^2 3x} = \lim_{x \rightarrow 0} \frac{x(1 - \cos x)}{\sin^2 3x} = \lim_{x \rightarrow 0} \frac{1}{3} \cdot \frac{3x}{\sin 3x} \cdot \frac{1 - \cos x}{\sin 3x} = \frac{1}{3} \cdot 1 \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{3 \sin x - 4 \sin^3 x} = \frac{1}{3} \cdot 1 \cdot \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{3 \cos \frac{x}{2} - 12 \sin^2 \frac{x}{2} \cos \frac{x}{2}} = \frac{1}{3} \cdot 1 \cdot \frac{0}{3-0} = 0$$

$$\frac{x(1-\cos x)}{\sin^2 3x} = \frac{(3x)^2}{\sin^2 3x} \cdot \frac{1-\cos x}{x} \cdot \frac{1}{9}$$

$$34. \lim_{h \rightarrow 0} \frac{\sin(\sin h)}{\sin h} = \lim_{k \rightarrow 0} \frac{\sin k}{k} = 1$$

* Let $\sin h = k$, $h \rightarrow 0$ means that $\sin h \rightarrow 0$ or $k \rightarrow 0$.

44. Yes, we can find the value of $\lim_{x \rightarrow c} f(x)$. Since we know that $\lim_{x \rightarrow c} f(x)$ exists when $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$. And $\lim_{x \rightarrow c} f(x)$ will be equal to those two values. If we could calculate the value of $\lim_{x \rightarrow c^-} f(x)$, that means we could refer its value to be $\lim_{x \rightarrow c} f(x)$.

45. Since $f(x)$ approaches 3 as x approaches 0 from the right and f is an odd function, we get that $f(x)$ approaches -3 as x approaches 0 from the left too. Hence, $\lim_{x \rightarrow 0} f(x) = -3$.

46. Since $f(x)$ approaches 7 as x approaches 2 from the left and f is an even function, it could be seen that as $f(x)$ approaches -2 from the right, $f(x)$ also approaches 7. However, we do not know if $\lim_{x \rightarrow 2^+} f(x)$ exists, so we do not know the value nor existence of $\lim_{x \rightarrow 2^-} f(x)$.